Dynamic EV Charging-Station Pricing: Problem Statement, Mathematical Model, and Numerical Solution

1 Problem statement

We consider a directed transportation network containing ordinary road links and charging-station links. Travellers belong to classes such as electric vehicles (EVs) and non-electric vehicles (NEVs). For every origin–destination (OD) pair, travellers choose among a bounded set of feasible routes. EV routes contain one reachable charging station, whereas NEV routes contain only road links.

The problem contains two coupled decision levels:

1. **Inner traffic and route-choice equilibrium.** For fixed charging prices, road occupancies, station occupancies, and route flows evolve according to a system of ordinary differential equations (ODEs). Travellers gradually move flow toward routes with lower perceived cost. The inner problem is solved until the complete ODE is close to a fixed point.

2. **Outer strategic pricing problem.** Every station changes its price to increase its own equilibrium profit. Changing one price changes travellers' route choices, congestion, station occupancy, and throughput; therefore profit must be evaluated after solving the inner equilibrium again.

For a price vector $\boldsymbol{\psi}$, let $\mathbf{z}^\star(\boldsymbol{\psi})$ denote the resulting inner equilibrium. Station s seeks to increase

$$\Pi_s(\boldsymbol{\psi}, \mathbf{z}^\star(\boldsymbol{\psi}))\,. \tag{1}$$

Thus the overall problem is a nested equilibrium problem rather than a single static shortest-path calculation.

2 Notation

Symbol	Meaning
$G = (V, E)$	Directed transportation graph, with node set V and link set E.
E_r	Set of ordinary road links.
E_s	Set of charging-station links.
$\mathcal{C}$	Set of vehicle classes, for example $\{\text{EV}, \text{NEV}\}$.
$\mathcal{W}$	Set of OD pairs.
$w = (o_w, d_w)$	OD pair with origin o_w and destination d_w.
$\lambda_w(t)$	Total exogenous demand rate for OD pair w at time t.
θ_{wc}	Share of OD demand belonging to class c, with $\sum_c \theta_{wc} = 1$.
$\lambda_{wc}(t)$	Class-specific demand, $\lambda_{wc}(t) = \theta_{wc}\lambda_w(t)$.

Symbol	Meaning
$\mathcal{P}_{wc}$	Bounded set of feasible paths for OD/class group (w, c).
p	A path in $\mathcal{P}_{wc}$.
δ_{ep}	Link–path incidence: 1 if link e belongs to path p, otherwise 0.
$x_{ec}(t)$	Occupancy/state of road link $e \in E_r$ for class c.
$x_s(t)$	Occupancy/state of charging station $s \in E_s$.
$y_p(t)$	Replicator route-flow state associated with path p.
$S_{wc}(t)$	Current route-flow total, $S_{wc}(t) = \sum_{p\in\mathcal{P}_{wc}} y_p(t)$.
$q_p(t)$	Physical demand assigned to path p.
$u_{ec}(t)$	Arrival rate to road link e for class c.
$u_s(t)$	Arrival rate to charging station s.
$f_{ec}(t)$	Road outflow rate from link e for class c.
$\rho_s(t)$	Charging-station throughput.
a_e	Linear road discharge coefficient.
L_e	Road flow/capacity scale used by the latency function.
ℓ_e^0	Road latency coefficient.
a_s	Unsaturated station discharge coefficient.
μ_s	Maximum station service rate.
K_s	Station saturation occupancy, $K_s = \mu_s/a_s$.
$W_s(x_s)$	Station waiting-time function.
ϕ_s^0	Fixed non-price component of station cost.
α	Weight assigned to station waiting time.
γ	Weight converting monetary price into perceived generalized cost.
ψ_s	Price charged by station s.
c_s	Per-vehicle operating cost of station s.
C_e	Generalized cost of link e.
C_p	Generalized cost of path p.
$\overline{C}_{wc}$	Current flow-weighted average path cost for group (w, c).
η	Route-choice/replicator adaptation rate.
$\mathbf{x}$	Vector containing every road and station state.
$\mathbf{y}$	Vector containing every multi-route replicator state.
$\mathbf{z}$	Complete inner state, $\mathbf{z} = [\mathbf{x}^\mathsf{T}, \mathbf{y}^\mathsf{T}]^\mathsf{T}$.
F	Complete ODE right-hand side, $\dot{\mathbf{z}} = F(\mathbf{z}; \boldsymbol{\psi})$.
R	Inner equilibrium residual, $R = \lVert\dot{\mathbf{x}}\rVert_2 + \lVert\dot{\mathbf{y}}\rVert_2$.
ε	Inner stopping tolerance; the implementation uses 10^{-6}.
$T_{\max}$	Maximum inner integration horizon; the default is 1000.
Π_s	Profit rate of station s.
Δ	Finite-difference price perturbation; the default is 0.02.
g_s	Estimated derivative of station s's profit with respect to its price.
$g_{\max}$	Gradient clipping bound; the default is 5.
κ	Outer price adaptation gain; the default is 0.1.
$\Delta\tau$	Outer-loop step duration; the default is 1.
$\psi_{\max}$	Maximum permitted station price; the default is 3.
k	Outer pricing-iteration index.

3 Step 1: feasible-route construction

For each group (w, c), the implementation first removes links that cannot be used by class c. It then constructs a bounded set $\mathcal{P}_{wc}$ of short, simple, feasible alternatives. NEV paths contain no station link. EV paths contain exactly one station link and are formed from a road path from the origin to a station, the station link itself, and a road path from the station to the destination.

Bounding $|\mathcal{P}_{wc}|$ is essential because enumerating every simple path can grow exponentially with network size. In preview mode, one best road prefix and suffix is retained per reachable station, followed by a limit of eight paths per OD/class group. Class-filtered road graphs and repeated path segments are cached during construction. The route set is part of the model: changing it can change the computed equilibrium.

4 Step 2: class-specific demand and route flow

The demand of class c for OD pair w is

$$\lambda_{wc}(t) = \theta_{wc}\lambda_w(t). \tag{2}$$

For a group with multiple paths, the physical flow assigned to path p is

$$q_p(t) = \lambda_{wc}(t)\frac{y_p(t)}{S_{wc}(t)}, \qquad S_{wc}(t) = \sum_{r\in\mathcal{P}_{wc}} y_r(t). \tag{3}$$

If there is only one feasible path, all class demand is assigned to it. If S_{wc} is numerically zero, equal route fractions are used defensively. Because the solver projects route flows onto

$$\mathcal{Y}_{wc} = \left\{ \mathbf{y}_{wc} \geq 0 : \sum_{p\in\mathcal{P}_{wc}} y_p = \lambda_{wc} \right\}, \tag{4}$$

normally $S_{wc} = \lambda_{wc}$ and hence $q_p = y_p$.

5 Step 3: road and station dynamics

For road link e and class c, the outflow is

$$f_{ec}(t) = a_e x_{ec}(t). \tag{5}$$

The road conservation equation is

$$\dot{x}_{ec}(t) = u_{ec}(t) - f_{ec}(t). \tag{6}$$

For charging station s, throughput is saturated:

$$\rho_s(x_s) = \min\{a_s x_s, \mu_s\}, \tag{7}$$

and station occupancy follows

$$\dot{x}_s(t) = u_s(t) - \rho_s(x_s(t)). \tag{8}$$

At an origin, the incoming flow to the first link of every route is q_p. At an internal link, the outgoing flow is divided among successor links using the route-flow proportions. If e is followed by j for some class-c paths, the implemented turning fraction can be written as

$$\tau_{e\to j,c} = \frac{\sum_{p:\,(e,j)\in p,\, p\in\mathcal{P}_{wc}} q_p}{\sum_{p:\, e\in p,\, p\in\mathcal{P}_{wc}} q_p}. \tag{9}$$

Consequently, $f_{ec}\tau_{e\to j,c}$ contributes to the arrival rate of successor j. This construction conserves flow through the network.

6 Step 4: generalized link and path costs

Let total occupancy on road e be

$$x_e^{\text{tot}} = \sum_{c \in \mathcal{C}} x_{ec}, \tag{10}$$

and let total road outflow be $f_e = a_e x_e^{\text{tot}}$. The implemented road cost is

$$C_e^r(x_e^{\text{tot}}) = \ell_e^0 \frac{f_e/L_e}{1 - f_e/L_e}, \qquad 0 \le \frac{f_e}{L_e} < 1. \tag{11}$$

Numerically, f_e/L_e is clipped below 1 to avoid division by zero.

The saturation occupancy and waiting time at station s are

$$K_s = \frac{\mu_s}{a_s}, \tag{12}$$

$$W_s(x_s) = \frac{\max\{x_s - K_s, 0\}}{\mu_s}. \tag{13}$$

The station's generalized user cost is

$$C_s^{\text{ch}}(x_s, \psi_s) = \phi_s^0 + \alpha W_s(x_s) + \gamma \psi_s. \tag{14}$$

The total cost of path p is additive:

$$C_p(\mathbf{x}, \boldsymbol{\psi}) = \sum_{e \in p \cap E_r} C_e^r + \sum_{s \in p \cap E_s} C_s^{\text{ch}}. \tag{15}$$

7 Step 5: route-choice dynamics

For each multi-route group (w, c), define its current flow-weighted average cost as

$$\overline{C}_{wc} = \frac{\sum_{p \in \mathcal{P}_{wc}} y_p C_p}{S_{wc}}, \qquad S_{wc} = \sum_{p \in \mathcal{P}_{wc}} y_p. \tag{16}$$

The route-flow state evolves according to replicator dynamics:

$$\dot{y}_p = \eta y_p \left(\overline{C}_{wc} - C_p \right), \qquad p \in \mathcal{P}_{wc}. \tag{17}$$

If $C_p < \overline{C}_{wc}$, then $\dot{y}_p > 0$ and flow moves toward path p. If $C_p > \overline{C}_{wc}$, then $\dot{y}_p < 0$.

The current-total denominator in Equation (16) preserves the group total:

$$\frac{\mathrm{d}S_{wc}}{\mathrm{d}t} = \sum_p \dot{y}_p \tag{18}$$

$$= \eta \left(\overline{C}_{wc} \sum_p y_p - \sum_p y_p C_p \right) \tag{19}$$

$$= \eta \left(\frac{\sum_p y_p C_p}{S_{wc}} S_{wc} - \sum_p y_p C_p \right) = 0. \tag{20}$$

This identity is the reason for normalizing by the current sum S_{wc} rather than by the target demand. It prevents a small numerical deviation in the route-flow total from being amplified by the vector field.

At a route-choice equilibrium, every used path has the same minimum cost:

$$y_p^\star > 0 \quad \Longrightarrow \quad C_p(\mathbf{x}^\star, \boldsymbol{\psi}) = \overline{C}_{wc}^\star. \tag{21}$$

This is the Wardrop-type user-equilibrium interpretation of the replicator fixed point.

8 Step 6: inner equilibrium computation

Collect all states into

$$\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \qquad \dot{\mathbf{z}} = F(\mathbf{z}; \boldsymbol{\psi}). \tag{22}$$

For fixed prices, an exact equilibrium satisfies

$$F(\mathbf{z}^\star; \boldsymbol{\psi}) = \mathbf{0}. \tag{23}$$

The implementation accepts a numerical equilibrium when

$$R(\mathbf{z}; \boldsymbol{\psi}) = \|\dot{\mathbf{x}}\|_2 + \|\dot{\mathbf{y}}\|_2 \le \varepsilon, \qquad \varepsilon = 10^{-6}. \tag{24}$$

The ODE is integrated with SciPy's implicit BDF method because the coupled traffic and replicator dynamics can be stiff. The default numerical settings are relative tolerance 10^{-7}, absolute tolerance 10^{-9}, maximum BDF step 15, chunk length 50, and maximum horizon $T_{\max} = 1000$.

For equilibrium integration only, the route-choice derivative is multiplied by a positive time-scale factor 20. This accelerates slow adaptation caused by the small price perturbations used for finite differences. Because $20\dot{\mathbf{y}} = \mathbf{0}$ if and only if $\dot{\mathbf{y}} = \mathbf{0}$, the equilibrium points are unchanged. The stopping residual in Equation (24) is evaluated using the original unscaled dynamics. Ordinary time-trajectory simulation also retains the original configured value of η.

After each chunk, route flows are projected back onto their simplex. Define $\widetilde{y}_p = \max\{y_p, 0\}$. If $\sum_r \widetilde{y}_r > 0$, the projection is

$$y_p \leftarrow \lambda_{wc} \frac{\widetilde{y}_p}{\sum_{r \in \mathcal{P}_{wc}} \widetilde{y}_r}. \tag{25}$$

If every clipped value is zero, demand is distributed equally. The residual in Equation (24) is recomputed after projection; therefore an event detected on an unprojected trajectory is not accepted blindly.

The solver also uses a sparse Jacobian pattern, warm-starts nearby price profiles from their previous equilibrium, caches verified equilibria, and returns a non-convergence warning rather than representing a finite endpoint as an exact fixed point.

9 Step 7: station profit and strategic price update

At inner equilibrium, station s earns

$$\Pi_s(\boldsymbol{\psi}) = (\psi_s - c_s)\rho_s\left(x_s^\star(\boldsymbol{\psi})\right). \tag{26}$$

The dependence of equilibrium throughput on every station price makes an analytic derivative inconvenient. The implementation uses a coordinate-wise central finite difference. For station s at iteration k, define

$$\psi_{s,k}^+ = \min\{\psi_{s,k} + \Delta, \psi_{\max}\}, \tag{27}$$

$$\psi_{s,k}^- = \max\{\psi_{s,k} - \Delta, 0\}. \tag{28}$$

All other station prices remain fixed. After separately solving the inner equilibrium at the two perturbed price vectors,

$$g_{s,k} \approx \frac{\Pi_s(\boldsymbol{\psi}_k^+) - \Pi_s(\boldsymbol{\psi}_k^-)}{\psi_{s,k}^+ - \psi_{s,k}^-}. \tag{29}$$

The estimate is clipped to $[-g_{\max}, g_{\max}]$, and projected gradient ascent gives

$$\psi_{s,k+1} = \mathrm{Proj}_{[0,\psi_{\max}]}\left(\psi_{s,k} + \Delta\tau\,\kappa\,\mathrm{clip}(g_{s,k}, -g_{\max}, g_{\max})\right). \tag{30}$$

The implementation allows at most $N = 30$ outer iterations by default and can stop earlier after the requested number of consecutive stable, fully converged iterations. With m stations, the upper bound on inner solves is

$$N(1 + 2m) + 1 \tag{31}$$

inner solves: one nominal and $2m$ perturbed solves per iteration, followed by one final solve. Independent perturbation solves are evaluated in parallel when worker processes are available.

10 Worked numerical example

Consider one EV OD pair with constant demand $\lambda = 1$ vehicle/time and two paths, p_1 through station s_1 and p_2 through station s_2. Assume

$$y_1 = 0.6, \qquad y_2 = 0.4, \qquad S = y_1 + y_2 = 1. \tag{32}$$

From Equation (3),

$$q_1 = 1\frac{0.6}{1} = 0.6, \qquad q_2 = 1\frac{0.4}{1} = 0.4. \tag{33}$$

Road costs

Let both roads have $\ell^0 = 0.2$, $L = 4$, and $a = 1$. Let their occupancies be $x_1 = 1$ and $x_2 = 0.5$. For road 1,

$$f_1 = ax_1 = 1, \tag{34}$$

$$C_1^r = 0.2\frac{1/4}{1 - 1/4} = 0.066667. \tag{35}$$

For road 2,

$$f_2 = ax_2 = 0.5, \tag{36}$$

$$C_2^r = 0.2\frac{0.5/4}{1 - 0.5/4} = 0.028571. \tag{37}$$

Station throughput, waiting time, and cost

Let both stations have $a_s = 0.5$, $\mu_s = 2$, $\phi_s^0 = 0.1$, $\alpha = 0.3$, and $\gamma = 1$. Thus

$$K_s = \frac{2}{0.5} = 4. \tag{38}$$

Suppose $x_{s_1} = 5$, $\psi_{s_1} = 0.5$, $x_{s_2} = 2$, and $\psi_{s_2} = 0.7$. Then

$$\rho_{s_1} = \min\{0.5(5), 2\} = 2, \tag{39}$$

$$W_{s_1} = \frac{\max\{5 - 4, 0\}}{2} = 0.5, \tag{40}$$

$$C_{s_1}^{\mathrm{ch}} = 0.1 + 0.3(0.5) + 1(0.5) = 0.75, \tag{41}$$

whereas

$$\rho_{s_2} = \min\{0.5(2), 2\} = 1, \tag{42}$$

$$W_{s_2} = 0, \tag{43}$$

$$C_{s_2}^{\mathrm{ch}} = 0.1 + 0 + 1(0.7) = 0.8. \tag{44}$$

Path costs and route adaptation

Each path contains its corresponding road and station, so

$$C_{p_1} = 0.066667 + 0.75 = 0.816667, \tag{45}$$
$$C_{p_2} = 0.028571 + 0.8 = 0.828571. \tag{46}$$

The current average cost is

$$\overline{C} = 0.6(0.816667) + 0.4(0.828571) \tag{47}$$
$$= 0.821429. \tag{48}$$

With $\eta = 0.05$, Equation (17) gives

$$\dot{y}_1 = 0.05(0.6)(0.821429 - 0.816667) = 1.4286 \times 10^{-4}, \tag{49}$$
$$\dot{y}_2 = 0.05(0.4)(0.821429 - 0.828571) = -1.4284 \times 10^{-4}. \tag{50}$$

The cheaper path gains flow and the more expensive path loses flow. Apart from rounding,

$$\dot{y}_1 + \dot{y}_2 = 0, \tag{51}$$

so total OD/class demand is preserved.

Physical-state derivative and equilibrium residual

Assume each path has a unique road followed by its station. Road arrivals are 0.6 and 0.4, while road outflows are 1 and 0.5. Therefore

$$\dot{x}_1 = 0.6 - 1 = -0.4, \tag{52}$$
$$\dot{x}_2 = 0.4 - 0.5 = -0.1. \tag{53}$$

The road outflows become station arrivals, so

$$\dot{x}_{s_1} = 1 - 2 = -1, \tag{54}$$
$$\dot{x}_{s_2} = 0.5 - 1 = -0.5. \tag{55}$$

Thus

$$\|\dot{\mathbf{x}}\|_2 = \sqrt{(-0.4)^2 + (-0.1)^2 + (-1)^2 + (-0.5)^2} \tag{56}$$
$$= \sqrt{1.42} = 1.19164, \tag{57}$$
$$\|\dot{\mathbf{y}}\|_2 \approx \sqrt{(1.4286 \times 10^{-4})^2 + (-1.4284 \times 10^{-4})^2} \tag{58}$$
$$\approx 2.020 \times 10^{-4}, \tag{59}$$
$$R \approx 1.19184. \tag{60}$$

Since $R \gg 10^{-6}$, this state is not an equilibrium and BDF continues integrating. At the accepted equilibrium, road and station inflows balance their outflows and used path costs are equal, making both derivative norms smaller than the required combined tolerance.

Profit and one price update

Let operating cost be $c_{s_1} = 0.2$. At the displayed state,

$$\Pi_{s_1} = (0.5 - 0.2)(2) = 0.6. \tag{61}$$

For the actual gradient, the inner equilibrium must be solved again at both perturbed prices. As an illustrative finite-difference step, let $\Delta = 0.02$ and suppose those two equilibrium solves produce

$$\rho^+_{s_1} = 1.9 \qquad \text{at } \psi^+_{s_1} = 0.52, \tag{62}$$
$$\rho^-_{s_1} = 2.1 \qquad \text{at } \psi^-_{s_1} = 0.48. \tag{63}$$

Then

$$\Pi^+_{s_1} = (0.52 - 0.2)(1.9) = 0.608, \tag{64}$$
$$\Pi^-_{s_1} = (0.48 - 0.2)(2.1) = 0.588, \tag{65}$$
$$g_{s_1} \approx \frac{0.608 - 0.588}{0.52 - 0.48} = 0.5. \tag{66}$$

Using $\kappa = 0.1$ and $\Delta\tau = 1$,

$$\psi_{s_1,\text{new}} = \text{Proj}_{[0,3]}\big(0.5 + 1(0.1)(0.5)\big) = 0.55. \tag{67}$$

Thus the station raises its price because its estimated equilibrium-profit gradient is positive.

11 Algorithm summary

1. Read the directed roads, stations, OD demands, vehicle classes, and model parameters.
2. Generate a bounded feasible-route set for every OD/class group.
3. Initialize road and station states to zero and distribute each multi-route demand equally across its paths.
4. Fix the current station-price vector $\boldsymbol{\psi}$.
5. Integrate the coupled traffic, station, and replicator ODE with BDF. After every chunk, project route flows onto their demand simplex.
6. Stop when $\|\dot{\mathbf{x}}\|_2 + \|\dot{\mathbf{y}}\|_2 \le 10^{-6}$ or report that $T_{\max}$ was reached without full convergence.
7. Compute station throughput and profit at the returned equilibrium.
8. For every station, perturb its price upward and downward and repeat the inner solve to estimate the coordinate profit gradient.
9. Apply the clipped, projected gradient-ascent price update.
10. Repeat until the requested outer-step limit is reached or prices remain stable for the requested number of converged iterations, then perform one final equilibrium solve at the final prices.

12 What the final result means

The output is an approximate coupled equilibrium. On the user side, no used route has an incentive to move flow to a more expensive route. On the station side, the final price vector is the endpoint of projected profit-gradient updates and is interpreted as an approximate local strategic-price equilibrium when price changes are small. A result is numerically trustworthy only when all required inner solves satisfy the residual tolerance, route-flow conservation error is negligible, and no important route set was truncated.